

ANSWER KEY

WORK SHEET – 04

1.

Particle	Mass No.	Atomic No.	Protons	Neutrons	Electrons
Nitrogen atom	14	7	7	7	7
Calcium ion	40	20	20	20	18
$^{80}_{35}\text{Br}$	80	55	35	45	35
$^{37}\text{Cl}^-$	37	17	17	20	18
Na^+	23	11	11	12	10

2. $^{32}_{16}\text{S}^{2-}$.

3. $^{40}_{18}\text{Ar}$, $^{40}_{19}\text{K}$, $^{40}_{20}\text{Ca}$

4. O^{2-} , F^- , Na^+ , Mg^{2+} , Al^{3+} , Ne .

5. $^{15}\text{N}_7$, $^{16}\text{O}_8$

WORK SHEET – 05

Ans.1 to 3

Q.No.	Frequency	Wavelength	Energy of one photon	Wave number
1	$7.5 \times 10^{14} \text{ Hz}$	400 nm	3.1 eV	$2.5 \times 10^6 \text{ m}^{-1}$
2	$3 \times 10^{14} \text{ Hz}$	10^{-6} m	$19.8 \times 10^{-17} \text{ kJ}$	10^6 m^{-1}
3	$4.8 \times 10^{14} \text{ sec}^{-1}$	6200 Å	2 eV	$\frac{10^8}{62} \text{ m}^{-1}$

4. 10^{21}

5. 2×10^{21}

WORK SHEET - 06

Ans. 1 to 3

Q.No.	Mass of particle	Velocity of particle	de-Broglie wavelength
1	0.1 kg	10 m/s	$6.626 \times 10^{-34} \text{ m}$
2	$0.5 \times 10^{-30} \text{ kg}$	10^6 m/s	13.2 Å
3	$3 \times 10^{-31} \text{ g}$	10^7 m/s	220 nm

4. 10 Å

5. $\sqrt{15} \text{ Å}$

WORK SHEET – 07

Ans.1

Q.No.	ν_0 (threshold)	ν (incident)	Maximum Kinetic energy
1	$6 \times 10^{14} \text{ Hz}$	$1.1 \times 10^{15} \text{ Hz}$	$3.313 \times 10^{-19} \text{ J}$
2	$4 \times 10^{10} \text{ Hz}$	$6 \times 10^9 \text{ Hz}$	0
3	$7 \times 10^{14} \text{ Hz}$	$1 \times 10^{15} \text{ Hz}$	$1.988 \times 10^{-19} \text{ J}$

Ans.2

Q.No.	Work function	λ (incident)	Maximum Kinetic energy
4	2.5 eV	310 nm	1.5 eV
5	$2.31 \times 10^5 \text{ J}$	300 nm	$1.68 \times 10^5 \text{ J}$

WORK SHEET – 08

Species	No. of orbit (n)	radius (in Å) of orbit	Velocity of electron(m/sec)	Total energy of electron (eV)	Pot. Energy of electron (eV)	Kinetic Energy of electron (eV)
He ⁺	2	1.058	2.18×10^6	-13.6	-27.2	13.6
Li ²⁺	3	1.587	2.18×10^6	-13.6	-27.2	13.6
Be ³⁺	2	0.529	4.36×10^6	-54.4	-108.8	54.4
H	2	2.116	1.09×10^6	-3.4	-6.8	3.4
He ⁺	4	4.232	1.09×10^6	-3.4	-6.8	3.4

WORK SHEET – 09

Series of H-Spectrum	Region	Longest Wavelength of series	Wavelength of Highest Energy Line	Series Limit (Wavelength)	Wavelength of n th line
Lyman	UV	4/(3R)	1/R	1/R	(3 rd line)
Balmer	Visible	36/(5R)	4/R	4/R	(5 th line)
Brackett series	Infrared	400/(9R)	16/R	16/R	(4 th line)
Pfund series	Infrared	900/(11R)	25/R	25/R	(2 nd line)
Paschen	Infrared	144/(7R)	9/R	9/R	(2 nd line)

WORK SHEET – 10

Ans.1

Q.No.	Ionization energy	2 nd excitation energy	Binding energy of 2 nd state
1	Li ²⁺	108.81 eV	30.6 eV
2	He ⁺	48.36 eV	13.6 eV
3	H	12.09 eV	3.4 eV

Ans.2

Q.No.	Species	Transition	Energy of emitted photons	Wavelength of emitted photons
1	H-atom	2 → 1	10.2 eV	$\frac{4}{3R}$
2	He ⁺ ion	3 → 2	7.56 eV	$\frac{9}{5R}$

Ans.3

S.No.	Initial level	Final level	Maximum Number of spectral lines
1	7th excited state	2nd state	21
2	7th state	1st excited state	15

WORK SHEET – 11

1. The uncertainty in the speed is 2%, i.e., $45 \times \frac{2}{100} = 0.9 \text{ m s}^{-1}$.

Using the equation

$$\Delta x = \frac{h}{4\pi m \Delta v} = \frac{6.626 \times 10^{-34}}{4 \times 3.14 \times 40 \times 10^{-3} (0.9 \text{ m s}^{-1})} = 1.46 \times 10^{-33} \text{ m}$$

This is nearly $\sim 10^{18}$ times smaller than the diameter of a typical atomic nucleus. As mentioned earlier for large particles, the uncertainty principle sets no meaningful limit to the precision of measurements.

2. (C)

$$\Delta x = 2\Delta p$$

$$\text{Now } \Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$2\Delta p^2 \geq \frac{h}{4\pi} \quad 2(m\Delta v)^2 \geq \frac{h}{4\pi} ; \Delta v \geq \frac{1}{2m} \sqrt{\frac{h}{2\pi}}$$

WORK SHEET – 12

Species	Electronic configuration	Unpaired electron	Spin only magnetic moment
${}^6\text{C}$	$1s^2 2s^2 2p^2$	2	2.83
${}^8\text{O}^{2-}$	$1s^2 2s^2 2p^4$	2	2.83
${}^{21}\text{Sc}$	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$	1	1.73
${}^{26}\text{Fe}^{2+}$	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$	4	4.9
${}^{24}\text{Cr}$	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$	6	6.93
${}^{29}\text{Cu}$	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$	1	1.73

WORK SHEET – 13

- 1.

Orbital	Angular node	Radial node	Total node	Orbital angular momentum
2s	0	1	1	0
2p	1	0	1	$\sqrt{2} \frac{h}{2\pi}$
4p	1	2	3	$\sqrt{2} \frac{h}{2\pi}$
1s	0	0	0	0
4f	3	0	3	$\sqrt{12} \frac{h}{2\pi}$
6s	0	5	5	0

2.

n	l	M	s	Mark (T / F)
3	2	2	+1/2	T
2	0	0	-1/2	T
4	2	-1	0	F
4	3	4	-1/2	F

WORK SHEET – 14

1.

S.No.	Properties	Alpha	Beta	Gamma
1	Nature	Fast moving He nuclei	First moving electrons	High energy radiations
2	Notation	${}_2\text{He}^4$ or α	${}_{-1}\text{e}^0$ or β	γ or ${}^0_0\gamma$
3	Charge	2 unit (+ ve)	1 unit (- ve)	No charge
4	Velocity	1/10 of light	33% to 90% of light	Same as light waves
5	Nature of path	Straight line	Crooked	Waves
6	Mass g/particle	6.65×10^{-24}	9.11×10^{-28}	0
7	Electrical or magnetic field's influence	Deflected towards -ve pole	Deflected towards +ve pole	Not deflected

2.

	4n	4n + 1	4n + 2	4n + 3
Parent element	${}_{90}\text{Th}^{232}$	${}_{94}\text{Pu}^{241}$	${}_{92}\text{U}^{238}$	${}_{92}\text{U}^{235}$
Name of series	Thorium (Natural)	Neptunium (Artificial)	Uranium (Natural)	Actinium (Natural)
End product	${}_{82}\text{Pb}^{208}$	${}_{83}\text{Bi}^{209}$	${}_{82}\text{Pb}^{206}$	${}_{82}\text{Pb}^{207}$
Number of lost particles	$\alpha = 6$ $\beta = 4$	$\alpha = 8$ $\beta = 5$	$\alpha = 8$ $\beta = 6$	$\alpha = 7$ $\beta = 4$